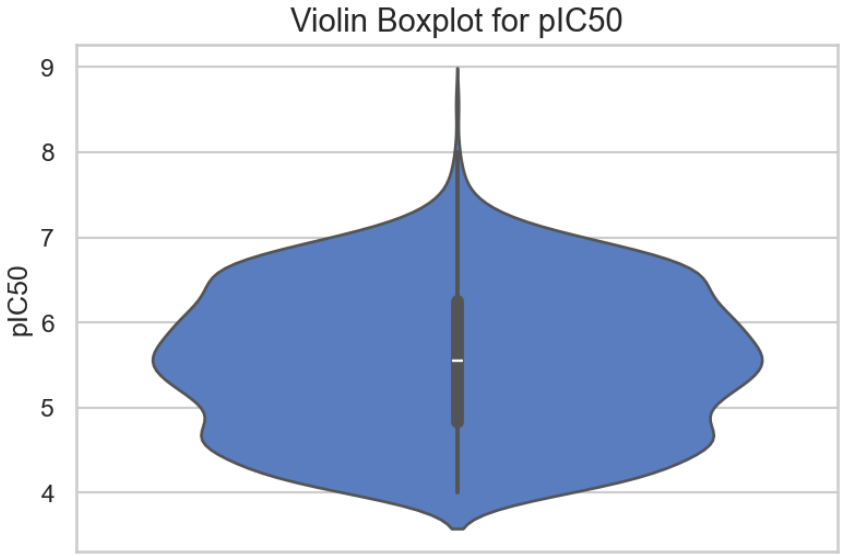
# Comprehensive Analytical Report for Mpro_continuous_report

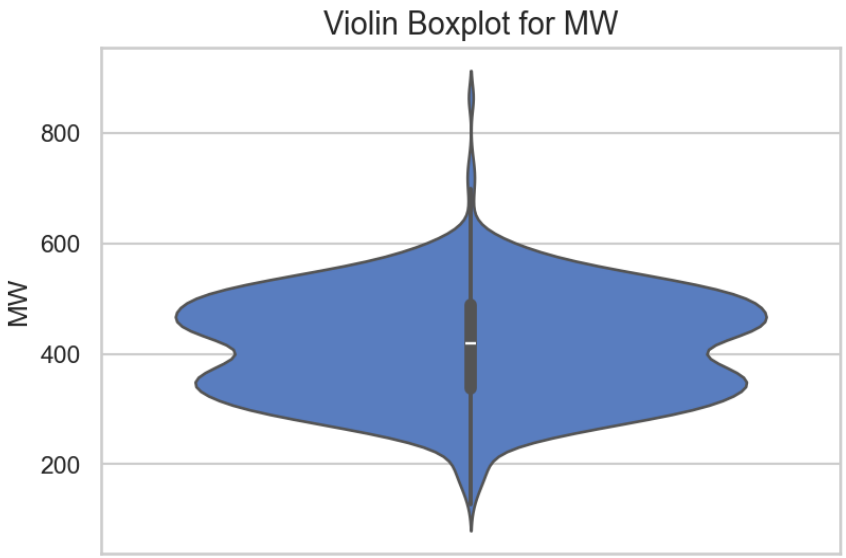
## Basic Statistics

| index | pIC50 | MW | LogP | HBD | HBA | TPSA | NumRotatableBonds | RingCount | FractionCSP3 | Heavy Atom Count | ESOL_LogS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| count | 1190.00 | 1190.00 | 1190.00 | 1190.00 | 1190.00 | 1190.00 | 1190.00 | 1190.00 | 1190.00 | 1190.00 | 1190.00 |
| mean | 5.55 | 414.12 | 3.59 | 1.38 | 4.55 | 80.17 | 4.86 | 3.80 | 0.24 | 28.95 | -5.73 |
| std | 0.87 | 96.88 | 1.24 | 1.09 | 1.82 | 32.62 | 2.49 | 1.16 | 0.13 | 7.07 | 2.27 |
| min | 4.00 | 127.01 | -2.71 | 0.00 | 1.00 | 3.24 | 0.00 | 0.00 | 0.00 | 8.00 | -23.50 |
| 25% | 4.83 | 338.09 | 2.99 | 1.00 | 3.00 | 57.69 | 3.00 | 3.00 | 0.16 | 24.00 | -7.19 |
| 50% | 5.55 | 419.12 | 3.69 | 1.00 | 4.00 | 79.37 | 5.00 | 4.00 | 0.24 | 29.00 | -5.59 |
| 75% | 6.24 | 488.19 | 4.33 | 2.00 | 6.00 | 97.60 | 6.00 | 5.00 | 0.31 | 34.00 | -3.93 |
| max | 8.57 | 865.19 | 7.13 | 12.00 | 19.00 | 363.63 | 23.00 | 8.00 | 1.00 | 62.00 | -0.72 |

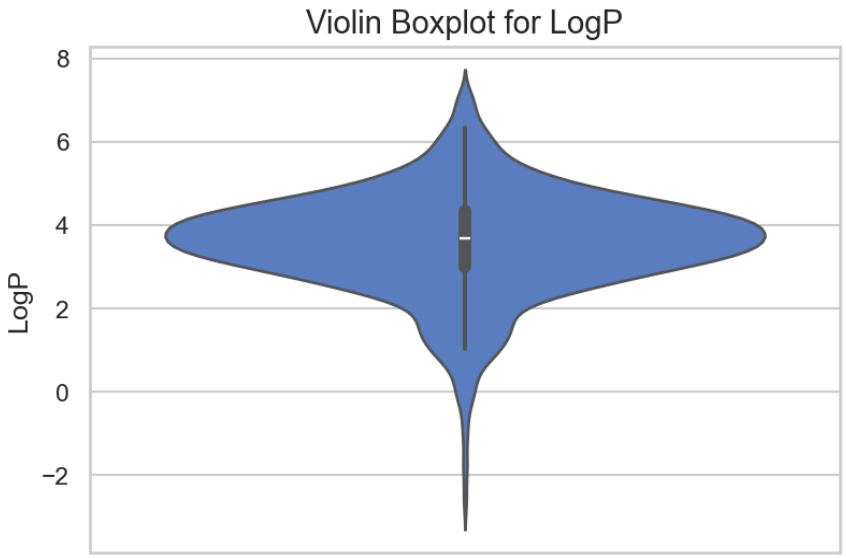
## Violin Boxplot for pIC50



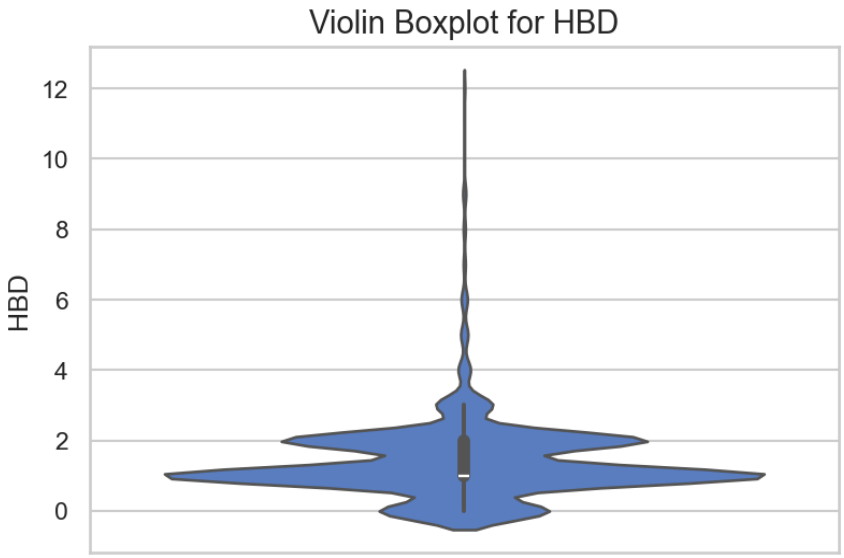
## Violin Boxplot for MW



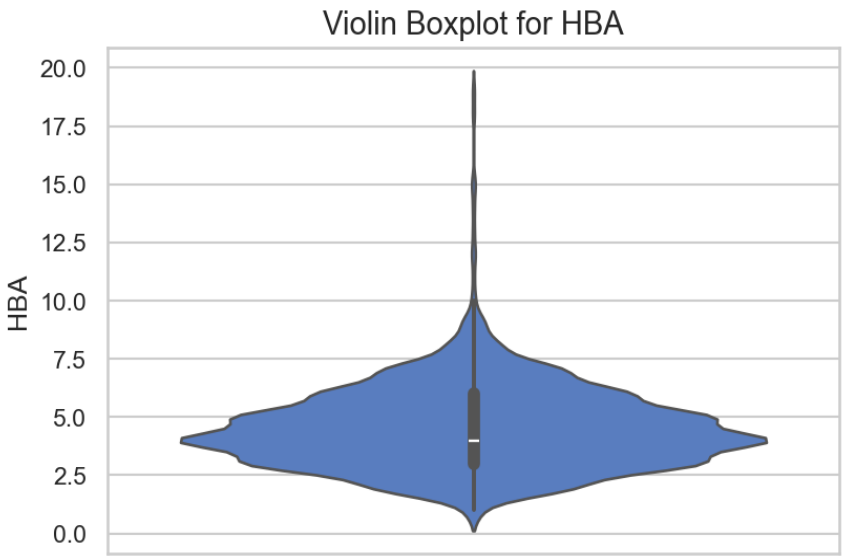

## Violin Boxplot for LogP



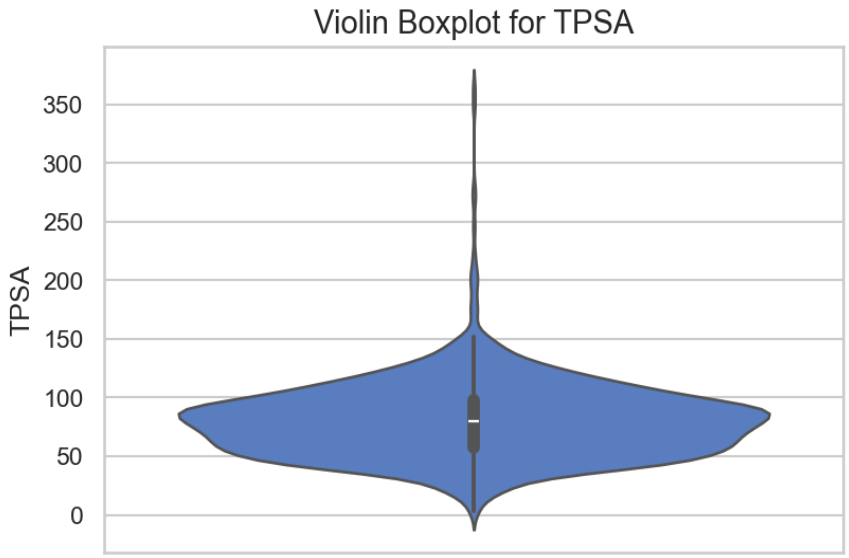

## Violin Boxplot for HBD



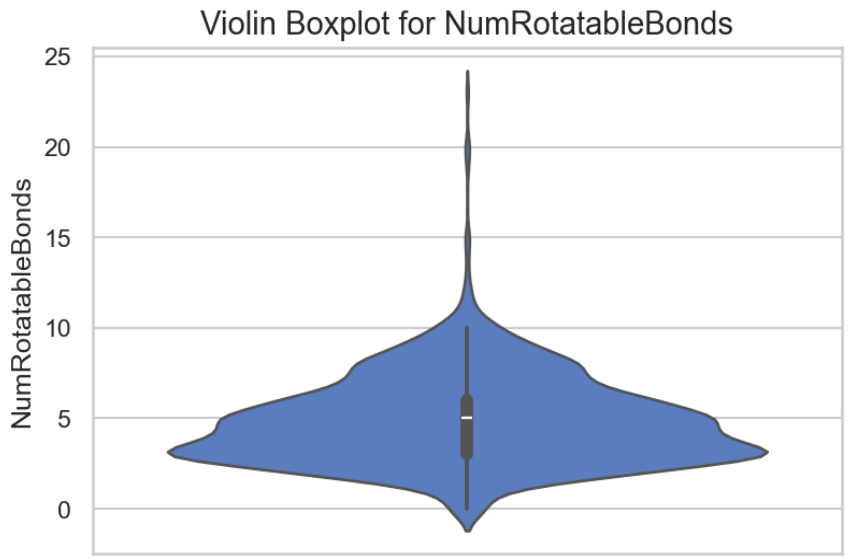

## Violin Boxplot for HBA



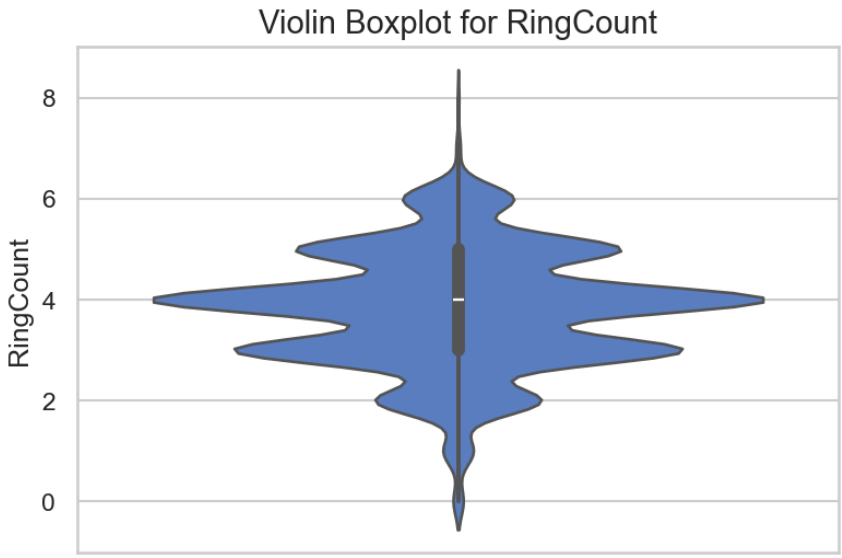

## Violin Boxplot for TPSA



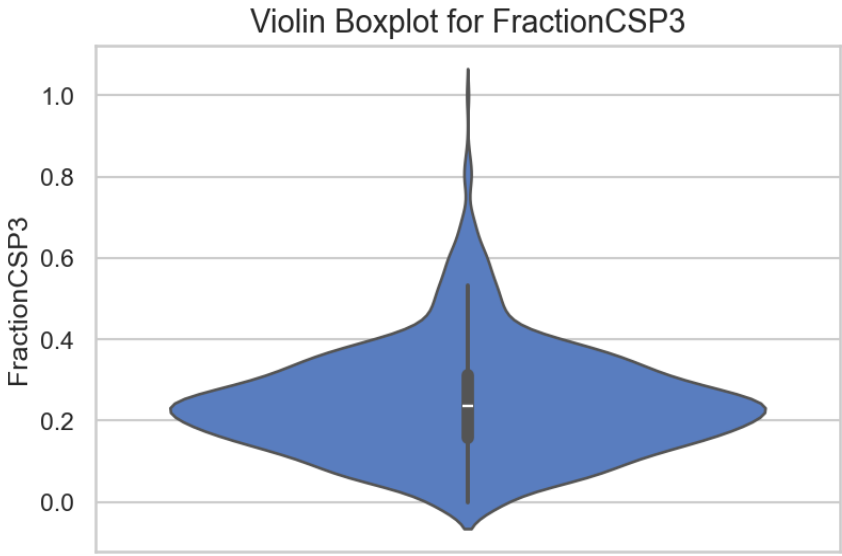

## Violin Boxplot for NumRotatableBonds



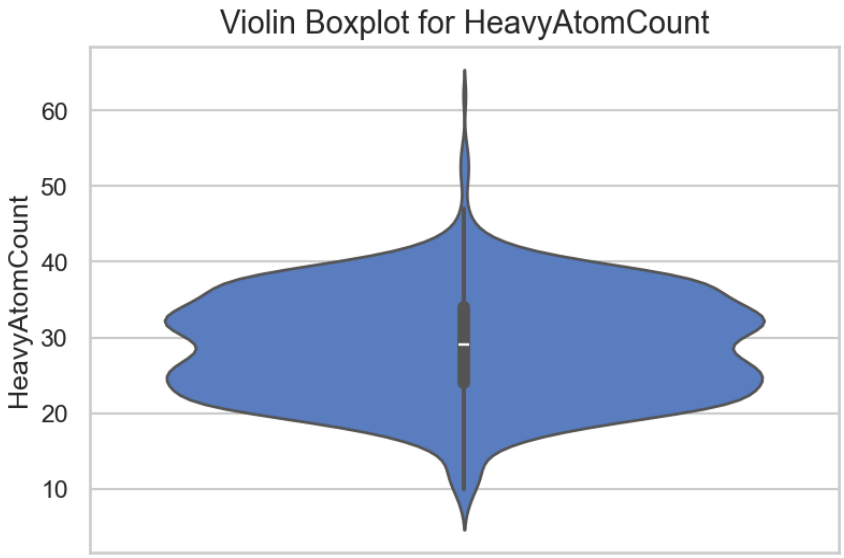

## Violin Boxplot for RingCount



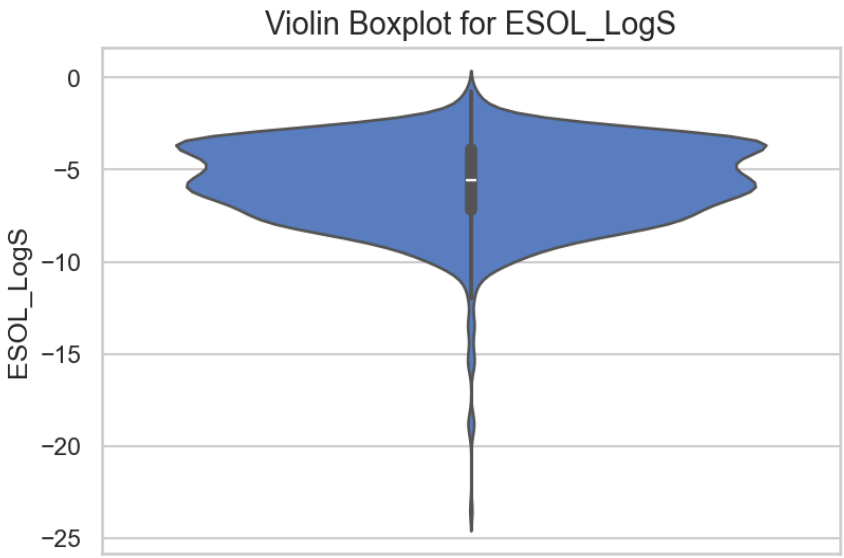

## Violin Boxplot for FractionCSP3



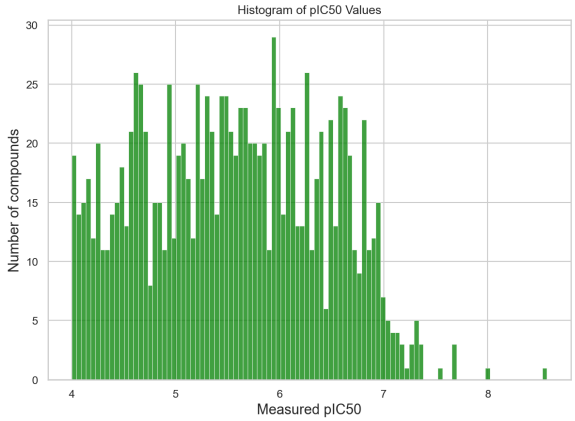

## Violin Boxplot for HeavyAtomCount

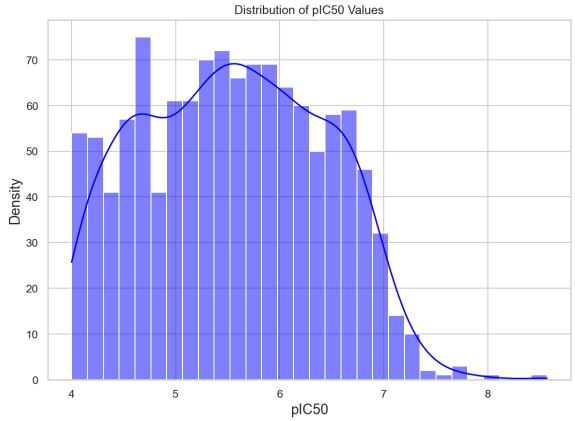


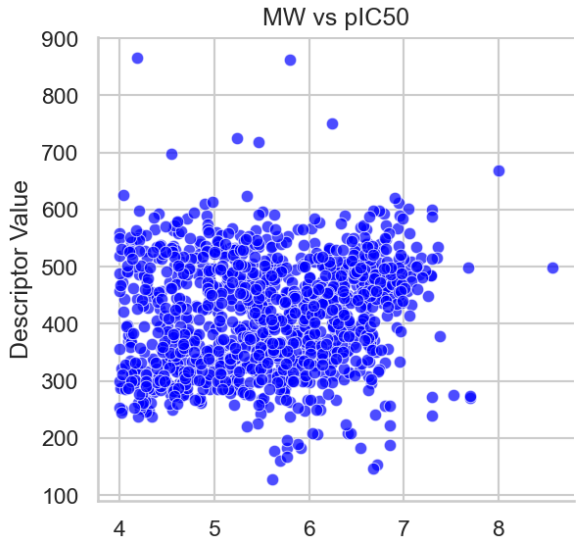

## Violin Boxplot for ESOL_LogS

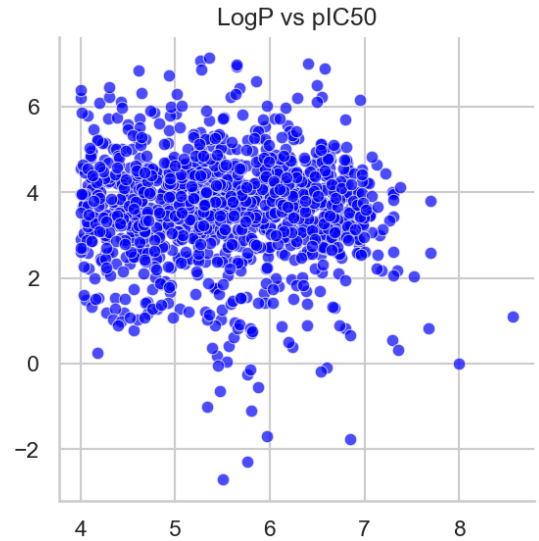


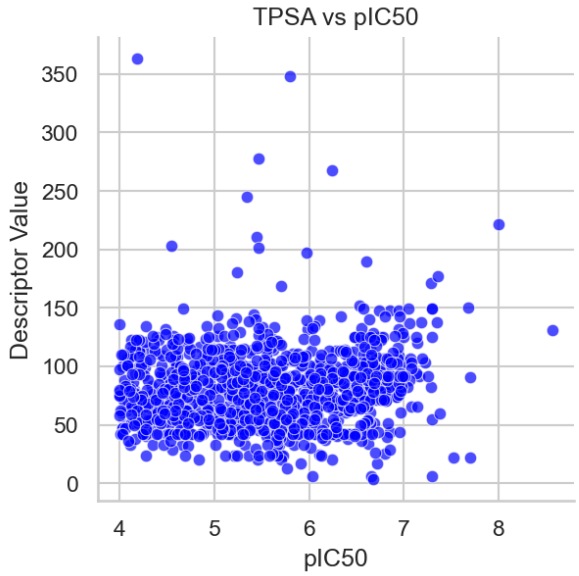

## Property Analysis

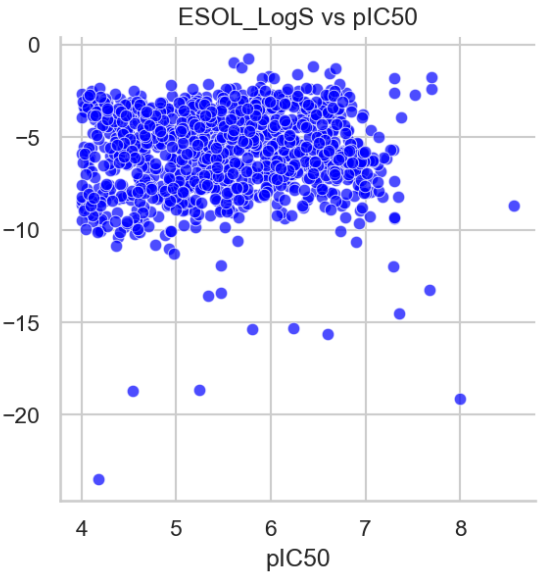

Histogram of pIC50 Values
Number of compounds
Measured pIC50

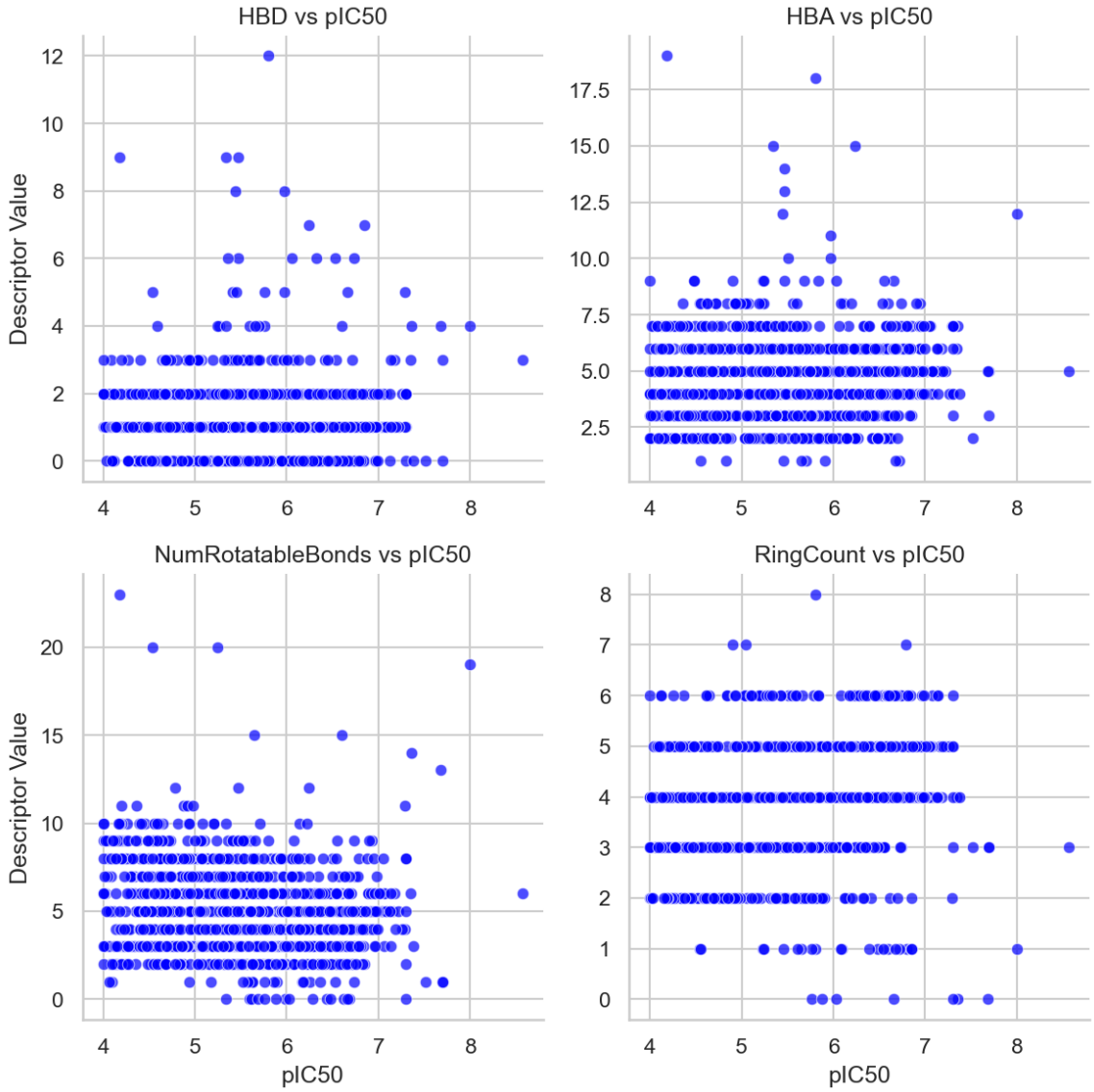

Distribution of pIC50 Values
Density
pIC50


MW vs pIC50
Descriptor Value


LogP vs pIC50

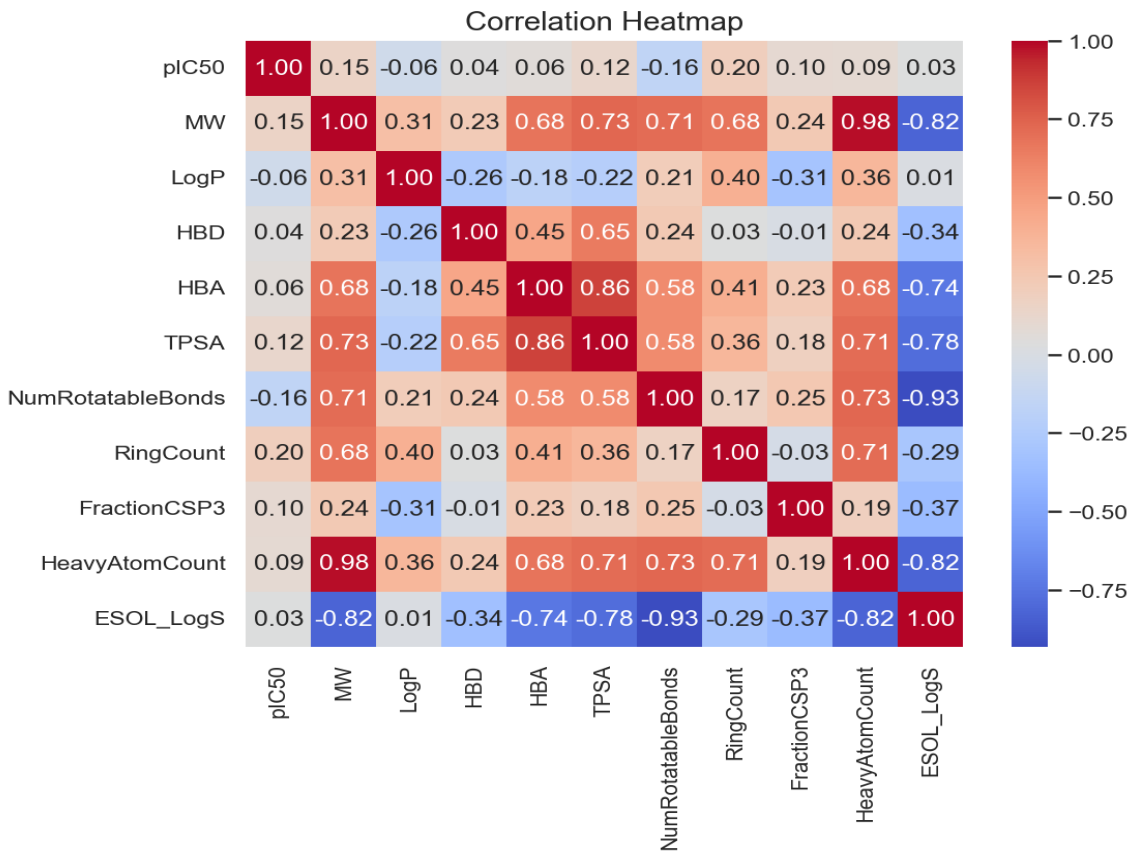

TPSA vs pIC50
Descriptor Value
pIC50


ESOL_LogS vs pIC50
pIC50




## Key Correlations

| | pIC50 | MW | LogP | HBD | HBA | TPSA | NumRotatableBonds | Ring Count | Fraction CSP3 | HeavyAtomCount | ESOL_LogS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| pIC50 | 1.0 | 0.15 | -0.06 | 0.04 | 0.06 | 0.12 | -0.16 | 0.2 | 0.1 | 0.09 | 0.03 |
| MW | 0.15 | 1.0 | 0.31 | 0.23 | 0.68 | 0.73 | 0.71 | 0.68 | 0.24 | 0.98 | -0.82 |
| LogP | -0.06 | 0.31 | 1.0 | -0.26 | -0.18 | -0.22 | 0.21 | 0.4 | -0.31 | 0.36 | 0.01 |
| HBD | 0.04 | 0.23 | -0.26 | 1.0 | 0.45 | 0.65 | 0.24 | 0.03 | -0.01 | 0.24 | -0.34 |
| HBA | 0.06 | 0.68 | -0.18 | 0.45 | 1.0 | 0.86 | 0.58 | 0.41 | 0.23 | 0.68 | -0.74 |
| TPSA | 0.12 | 0.73 | -0.22 | 0.65 | 0.86 | 1.0 | 0.58 | 0.36 | 0.18 | 0.71 | -0.78 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| NumRotatableBonds | -0.16 | 0.71 | 0.21 | 0.24 | 0.58 | 0.58 | 1.0 | 0.17 | 0.25 | 0.73 | -0.93 |
| RingCount | 0.2 | 0.68 | 0.4 | 0.03 | 0.41 | 0.36 | 0.17 | 1.0 | -0.03 | 0.71 | -0.29 |
| FractionCSP3 | 0.1 | 0.24 | -0.31 | -0.01 | 0.23 | 0.18 | 0.25 | -0.03 | 1.0 | 0.19 | -0.37 |
| HeavyAtomCount | 0.09 | 0.98 | 0.36 | 0.24 | 0.68 | 0.71 | 0.73 | 0.71 | 0.19 | 1.0 | -0.82 |
| ESOL_LogS | 0.03 | -0.82 | 0.01 | -0.34 | -0.74 | -0.78 | -0.93 | -0.29 | -0.37 | -0.82 | 1.0 |

## Correlation Heatmap